annealing temperature, emission lifetime (τ), and operational lifetime (OLT). The tabulated answer is in Table 14.

Table 15: Answer table for Q.17 joining Tables 3 and 5 of Ghazy et al. [25], modeled in the ORKG as Comparisons R1471077 (Table 3) and R1469991 (Table 5). For each luminescent MOSLED host matrix, the table lists EQE together with the aligned ALD process parameters: process material, metal precursor, co-reactant, GPC, and deposition temperature.

Matrix	**Process material**	**EQE [%]**	**Metal precursor**	**Co-reactant**	**GPC [Å/cycle]**	**T [°C]**
$Lu_3Al_5O_{12}$	$Lu_3Al_5O_{12}$	10.0	$Lu(thd)_3$	TMA	0.19	350
Yb_2O_3	Yb_2O_3	8.5	$Yb(thd)_3$	O_3	0.15–0.20	250–400
Yb_2O_3	Yb_2O_3	8.5	$Yb(thd)_3$	O_3	0.15–0.20	250–400
Yb_2O_3	Yb_2O_3	8.5	$Yb(thd)_3$	O_3	0.15–0.20	250–400
Yb_2O_3	Yb_2O_3	8.5	$Yb(MeCp)_3$	H_2O	1.00	180–375
$Yb_3Al_5O_{12}$	$Yb_3Al_5O_{12}$	5.2	$Yb(thd)_3$	TMA	0.20	350
$Y_3Ga_5O_{12}$	$Y_3Ga_5O_{12}$	2.5	$Y(thd)_3$	Et_3Ga	0.23	350
$Gd_3Ga_5O_{12}$	$Gd_3Ga_5O_{12}$	1.9	$Gd(thd)_3$	TEGa	0.19	350

Complex Query 1. The seventeenth question is: *"For each luminescent MOSLED host material listed in the rare-earth MOSLED performance comparison, retrieve the corresponding ALD process parameters from the rare-earth ALD process overview—metal precursor, co-reactant, growth-per-cycle rate, and deposition temperature—and report them alongside the external quantum efficiency (EQE)."* The SPARQL query implementation for this question can be rerun via `https://tinyurl.com/t3t5-complex` over the ORKG comparisons R1471077 (Table 3) and R1469991 (Table 5). The tabulated answer is in Table 15.

Table 16: Answer table for Q.18 joining Tables 3 and 5 of Ghazy et al. [25], modeled in the ORKG as Comparisons R1471077 (Table 3) and R1469991 (Table 5). For each matrix appearing in both tables, the synthesis temperature, annealing temperature, EQE, power efficiency (PE), and emission lifetime are listed.

Matrix	**Syn. Temp [°C]**	**Ann. Temp [°C]**	**EQE [%]**	**PE [%]**	**Lifetime [ms]**
Ga_2O_3	350	900	36.0	81.0	2.04
Al_2O_3:Yb	350		24.0	28.0	0.90
$Lu_3Al_5O_{12}$	350	1100	10.0	12.0	1.65
$Al_2O_3-Y_2O_3$	350	800	8.7	12.0	
Yb_2O_3	350	1000	8.5	10.0	
$Yb_3Al_5O_{12}$	350	1100	5.2	4.8	1.20
$Y_3Ga_5O_{12}$	350	800	2.5	3.8	2.26
$Gd_3Ga_5O_{12}$	350	1000	1.9	3.6	1.59

Complex Query 2. The eighteenth question is: *"Which rare-earth oxide matrices share process-level synthesis information and device-level performance data across Tables 3 and 5, and how do synthesis and annealing conditions relate to their optical efficiencies and lifetimes?"* The SPARQL query implementation for this question can be rerun via `https://tinyurl.com/t3t5-correlation` over the ORKG comparisons R1471077 (Table 3) and R1469991 (Table 5). For each matrix occurring in both tables, it retrieves the synthesis temperature, annealing temperature, external quantum efficiency (EQE), power efficiency (PE), and emission lifetime. The tabulated answer is in Table 16.

Complex Query 3. The nineteenth question is: *"From Table 2, select oxide materials doped with Er for luminescence. Combine their synthesis temperatures (Table 3) and EQEs (Table 5), then compute an efficiency index = (EQE / synthesis temperature $\times$ 100) to identify materials that deliver high efficiency at lower fabrication temperatures. Rank materials by this index."* The SPARQL query implementation for this question can be rerun via `https://tinyurl.com/t2t3t5-complex` over the ORKG comparisons R1469955 (Table 2), R1471077 (Table 3), and R1469991 (Table 5). The tabulated answer is in Table 17.

As a final note to this subsection, we publish all four machine-actionable ORKG comparisons derived from the tables of the Ghazy *et al.* (2025) review article *"Atomic and molecular layer*

Table 17: Answer table for Q.19 joining Tables 2, 3, and 5 of Ghazy et al. [25], modeled in the ORKG as Comparisons R1469955 (Table 2), R1471077 (Table 3), and R1469991 (Table 5). For Er-doped luminescent oxides appearing across all three tables, the synthesis temperature, EQE, and derived efficiency index (EQE/synthesis temperature × 100) are listed.

Material	**Syn. Temp [°C]**	**EQE [%]**	**Efficiency Index**
Ga_2O_3	250	36.0	14.4
Al_2O_3	250	24.0	9.6
Y_2O_3	140	8.7	6.21
Yb_2O_3	180	8.5	4.72
$Lu_3Al_5O_{12}$	230	10.0	4.35
$Yb_3Al_5O_{12}$	350	5.2	1.49
$Y_3Ga_5O_{12}$	350	2.5	0.71
$Gd_3Ga_5O_{12}$	350	1.9	0.54

deposition of functional thin films based on rare earth elements" [25]: Table 2 as `https://orkg.org/comparisons/R1469955`, Table 3 as `https://orkg.org/comparisons/R1471077`, Table 4 as `https://orkg.org/comparisons/R1470110`, and Table 5 as `https://orkg.org/comparisons/R1469991`. These four comparisons are jointly curated in the ORKG as a mini review article in the Smart Review format (`https://orkg.org/reviews/R1471098`). The Smart Review serves as a concise, machine-actionable counterpart to the original review, illustrating how structured, queryable representations of ALD data can complement and extend traditional scholarly publishing.

With this, we conclude the section on ALD machine-actionable comparisons in the ORKG. In total, seven tables from three recent ALD review papers were converted from their original PDF form into citable, machine-actionable ORKG Comparisons. Over these representations, we demonstrated the benefits of machine-actionable data by executing 19 SPARQL queries.

Next, we turn to Atomic Layer Etching (ALE). ALE—the subtractive analogue of ALD—uses sequential, self-limiting surface reactions to remove material with Å-level precision. Plasma-based ALE has become well established in semiconductor processing, while thermal ALE has advanced rapidly through mechanistic studies and expanding materials coverage [34, 35, 23, 20]. In the remainder of this section, we apply the same ORKG-based modeling and querying approach to representative ALE review tables.

To demonstrate machine-actionable modeling for ALE, we selected six review-style articles that each contain at least one substantive table comparing process conditions, materials, or etch outcomes across multiple studies and including a dedicated "Ref." or "References" column. Such tables align well with the ORKG publishing model, where each row is linked to its source article and each column corresponds to a well-defined property. Together, these papers cover both plasma and thermal ALE, from general overviews to materials-specific studies. In the following subsections, we convert their tables into ORKG Comparisons and illustrate Easy and Complex Queries over the reported etching parameters and outcomes.

3.4 Paper – Atomic layer etching at the tipping point: an overview

In this seminal review, Oehrlein *et al.* (2015) [43] provide an early comprehensive overview of ALE, outlining key concepts, mechanisms, and materials systems, and comparing halogen- and fluorine-based chemistries across oxides, III–V semiconductors, and related substrates. From this work, we model Table I (*"Overview of materials and ALE investigations"*) as a machine-actionable ORKG Comparison, capturing the surveyed ALE investigations in structured form as a starting point for an ALE knowledge graph.

3.4.1 Machine-actionable modeling of Table I

Based on the ALE survey summarized in Table I of Oehrlein *et al.* (2015) [43], we modeled the reported materials, chemistries, and energy sources as an ORKG Comparison `https://orkg.org/comparisons/R1562672` [21]. This machine-actionable comparison captures, for each ALE investigation, the etched material, the precursor chemistry used in the adsorption step, and the energy source driving etching or desorption, yielding a structured, queryable representation of the table.

The underlying data span oxides, compound semiconductors, nitrides, polymers, and carbon-based materials, illustrating the diversity of ALE mechanisms developed to achieve atomic-scale surface removal.

Table 18: Answer table for Q.20 over Table I of Oehrlein et al. [43], modeled in the ORKG as Comparison R1562672. The table lists, for each ALE investigation, the material, adsorption precursor chemistry, and energy source used.

Material	Precursor Chemistry	Energy Source
Al_2O_3	BCl_3	Ar neutral beam
BeO	BCl_3	Ar neutral beam
GaAs	Cl_2	Electron bombardment
GaAs	Cl_2	248 nm KrF excimer and Ti:sapphire lasers
InP	Tertiarybutylphosphine	Halogen lamp desorption
Ge	Cl_2	Ar ions from ECR plasma
Graphene	O_2 plasma	Ar neutral beam
HfO_2	BCl_3	Ar neutral beam
Polymer (polystyrene)	O_2	Ar ions from CCP plasma
Si	Cl_2	Ar ions from ICP source
TiO_2	BCl_3	Ar neutral beam

Easy Query. The twentieth question is: *"For each ALE investigation summarized in Table I, list the material, adsorption precursor chemistry, and energy source used for etching or desorption."* The SPARQL query implementation for this question can be rerun via `https://tinyurl.com/orkg-ale-materials-nolimit`. As explained earlier, answers in this work are tabulated, synthesized results based on the original question; Table 18 lists all materials, precursor chemistries, and energy sources recorded in the ORKG comparison.

Complex Query. The 21st question is: *"Group all ALE materials by the dominant energy-source category (neutral beam, plasma ions, photon, or thermal), and count how many distinct precursor chemistries are used within each category."* The SPARQL query implementation for this question can be rerun via `https://tinyurl.com/ale-energy-classes`. As with the other questions, the answer is a synthesized, machine-derived aggregation over the ORKG comparison: all ALE entries are grouped by energy-source category, and for each category the number of distinct precursor chemistries is returned. In the current comparison, plasma- or ion-driven ALE processes are associated with 12 distinct precursor chemistries, neutral-beam approaches with four, and photon-assisted and other activation schemes with one each.

3.5 Paper – Thermal atomic layer etching: A review

The review by Fischer *et al.* (2021) provides an overview of *thermal ALE*, focusing on self-limiting, plasmaless reaction mechanisms based on sequential surface fluorination and ligand-exchange steps. It surveys experimental and theoretical work across oxides, nitrides, semiconductors, and metals, and discusses underlying chemical principles, reactor designs, and mechanistic classifications. Among its contents, only Table III met our structural and provenance criteria for machine-actionable modeling in the ORKG, as it systematically lists ALE publications by etched material and reactant chemistry. We therefore converted Table III into a semantically structured ORKG comparison to enable reproducible, queryable synthesis of ALE chemistries across the literature.

3.5.1 Machine-actionable modeling of Table III

Table III of the review article *"Thermal Atomic Layer Etching: A Review"* lists ALE publications by etched material and reactant chemistry. Each row corresponds to one experimentally reported process, recording the substrate material, primary and secondary reactants, and bibliographic reference. In its machine-actionable form, this table is modeled in the ORKG as Comparison `https://orkg.org/comparisons/R1563034` [4], where each publication entry is represented as a `compareContribution` with four key predicates: P183144 (*material etched*), P183145 (*reactant 1*), P183146 (*reactant 2*), and P183147 (*reactant 3*). This encoding turns the literature summary into a semantically queryable dataset that underlies the Easy and Complex Queries discussed below.